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Delegation with Technology Migration: An Empirical Analysis of Mobile Virtual Network Operators

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Abstract. This study examines the impact of mobile virtual network operators (MVNOs) on the performance of mobile network operators (MNOs) in the presence of overlapping generations of wireless mobile network technologies (2G and 3G). MVNOs distribute MNOs' mobile services to customers without owning any spectrum or network infrastructures, either as a subsidiary (branded MVNOs) or delegated under wholesale agreements with MNOs (third-party MVNOs). We investigate MVNO value chain governance, that is, delegation, in the context of overlapping generations of technologies. Leveraging a proprietary data set, we design a quasi-experiment to analyze the impact of branded and third-party MVNOs in 2G and 3G market expansion. Using first a difference-in-differences analysis and then dynamic propensity score matching, we find that an MNO's 2G market share increases if the MNO launches a third-party MVNO, whereas its 3G market share increases if it launches a branded MVNO. More importantly, the positive effects of branded MVNOs diminish over time, but those of third-party MVNOs become stronger over time. For managers challenged by managing overlapping generations of technologies, it is important to customize value chain governance for market expansion: to create branded MVNOs to explore new mobile technologies, and to delegate to third-party MVNOs to exploit existing technology markets when the technology market is mature.

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Keywords: value chain governance • mobile network technology • dynamic propensity score matching

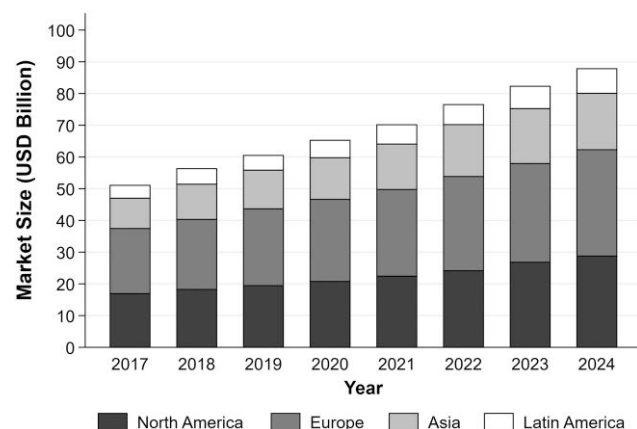
1. Introduction

In the telecommunication industry, mobile network operators (MNOs) adopt and adjust strategies with rapidly changing technologies and markets. One recent MNO strategy involves mobile virtual network operators (MVNOs) that provide mobile services to end-users without owning spectrum or network infrastructures (Shin 2010). For example, AT&T, one of the major MNOs in the United States, had twenty-four MVNOs, such as Cricket Wireless, by 2019. In fact, “mobile network operators are increasingly sharing the networks (with MVNOs) in order to lower costs, maintain profit margins and focus on innovative services to meet shifting customer demands” (Clark et al. 2017, p. 1). Specifically, MVNOs often target under-served niche markets (e.g., demographic groups) where customers tend to be more price-sensitive with low usage needs (FCC 2013, p. 3738). The fast expansion of MVNOs is projected to reach a total size of 89 billion U.S. dollars by 2024 as shown in Figure 1, which signifies the significant role of

MVNOs as a value chain strategy in market expansion (Tilson and Lyytinen 2006). In practice, however, this strategy has faced challenges of constantly changing technologies and markets with overlapping generations of mobile network technologies, which require agility and adaptability, and limited market scalability due to significant resource constraints, which highlights the need for operational efficiency. This study aims to examine value chain governance with MVNOs and overlapping generations of technologies in highly dynamic mobile network markets.

Breakthroughs in technology typically lead to innovative designs that compete against each other within and across generations of technology for dominance (Anderson and Tushman 1990). However, overlapping generations of technologies such as those found in the mobile network industry depend on industry collaboration to develop global standards and product designs. While the literature on dominant design has shed light on the competitive nature of design decisions, managing such

Figure 1. Global MVNO Market Size by Region from 2017 to 2024



overlapping generations of technologies involves the exploration and exploitation of technologies beyond product design (March 1991, Chandrasekaran et al. 2012). Exploitation of existing technology markets focuses on efficiently extracting the remaining value of the markets that have limited growth potential before being replaced by new technology (March 1991, O'Reilly and Tushman 2013). Exploration of new technology markets, on the other hand, deals with highly diverse, quickly shifting, and highly unpredictable demand, and therefore benefits from flexibility (O'Reilly and Tushman 2013, Lee et al. 2015). Effective management of overlapping generations of technologies, therefore, hinges on the trade-off between flexibility and efficiency (Adler et al. 1999, Ebben and Johnson 2005), a trade-off which is calibrated based on the types of MVNOs in the mobile network value chain.

MVNOs are introduced either as branded or third-party MVNOs (Lehikoinen et al. 2014). A branded MVNO is a subsidiary of its MNO, which functions as internal slack and requires additional MNO resource commitment for capacity and flexibility (Nohria and Gulati 1996, Cheng and Kesner 1997, Voss et al. 2008, Bentley and Kehoe 2020). Internal slack, however, underutilizes assets and resources, which reduces organizational efficiency (Jensen and Meckling 1976, Tan and Peng 2003). Alternatively, third-party MVNOs are independent entities in the form of value chain delegation that allows MNOs to leverage external, complementary resources, such as localized knowledge, specialized expertise, and brand recognition (Camarán and De Miguel 2008). By delegating, MNOs can quickly access underserved market segments (e.g., industrial, demographic, and geographical) that are difficult or costly to reach by themselves, thus improving organizational efficiency (Varoutas et al. 2006, Jaspers et al. 2007). However, delegation also weakens MNOs' flexibility in market expansion because third-party MVNOs are independent and driven solely by their own interests that may be

inconsistent with the interests of the MNOs (Adler et al. 1999, Ebben and Johnson 2005).

Prior research on MVNOs is scarce and largely focuses on MVNO success factors and on policies that impact MVNOs (Kim and Seol 2007, Shin and Bartolacci 2007, Lee et al. 2008, Corrocher and Lasio 2013). A broader stream of management research extensively examines the exploitation and exploration in intrafirm activities (March 1991, Levinthal and March 1993, O'Reilly and Tushman 2004, Raisch and Birkinshaw 2008, Jansen et al. 2009, Chandrasekaran et al. 2012, Kang and Kim 2020). The few interfirm studies focus on horizontal relationships of strategic alliance and mostly overlook the vertical relationships of value chain governance (e.g., Rothaermel and Deeds 2004, Lin et al. 2007, Lavie et al. 2011). On the other hand, literature on delegation indicates that delegation is positively associated with efficiency and negatively associated with uncertainty, which is relevant to the trade-off behind the exploration and exploitation of overlapping generations of technologies (Lee and Whang 2002, Wu 2004, Brahm and Tarzijań 2016). However, the delegation literature focuses on governing mechanisms, such as pricing contracts, and rarely provides relevant empirical evidence, particularly of delegation's role in the dynamic aspects of technologies. In this study, we investigate value chain delegation in the context of overlapping generations of mobile network technologies from the theoretical lens of exploitation and exploration. Specifically, we address the following empirical question: *In the presence of overlapping generations of technologies, should MNOs create branded MVNOs for new technologies and delegate to third-party MVNOs for existing technologies for more effective market expansion?*

In high-velocity environments such as the mobile network market, MNOs must integrate and reconfigure internal and external resources to adapt to technological and market dynamics (Teece et al. 1997, Chandrasekaran et al. 2012, Luger et al. 2018). From a theoretical standpoint, the exploitation of existing technologies becomes less efficient in the diffusion process of new technologies. By the same token, the governance structures designed to accommodate previous technologies may no longer be effective to embrace new technologies (Siggelkow and Levinthal 2003, Fuentelsaz et al. 2008, Kang and Kim 2020). The literature has not addressed value chain governance from this important dynamic perspective of overlapping generations of technology. We fill this literature gap by focusing on the following question: *Do the market expansion effects of creating branded MVNOs and delegating to third-party MVNOs change as mobile network technology markets mature?*

To address these research questions, we examine the market expansion effects of MVNOs with different value chain governance (i.e., branded and third-party) and different generations of mobile network technologies (2G

and 3G mobile network services). Using proprietary data, we employ a quasi-experimental design to analyze the impact of branded and third-party MVNOs on MNOs' market expansion in the 2G and 3G markets.¹ We address our first research question by employing a sliding-window-adjusted difference-in-differences (DID) model to test the average treatment effects. To properly address the dynamic effects in our second research question, we adopt dynamic propensity score matching (dPSM), a method designed to incorporate temporal heterogeneity of the treatment effects, therefore providing a view of the dynamic process during technology maturation.

We find that, first, the 2G market share of an MNO increases when the MNO launches a third-party MVNO. This indicates that independent MVNOs are an efficient value chain solution in market expansion with existing technologies. Delegating to a third-party MVNO provides the MNO with direct access to complementary resources (e.g., strong representation and know-how of niche market segments), allowing the MNO to benefit from the exploitation of existing technologies for efficiency (March 1991). Importantly, market expansion with third-party MVNOs in the 2G market accelerates over time, highlighting the increasing efficiency from complementary resources as the technology market matures. Second, the 3G market share of an MNO increases when the MNO launches a branded MVNO. This suggests that MNOs with new technologies indeed benefit from a more flexible environment for market expansion (March 1991, O'Reilly and Tushman 2013). Launching a branded MVNO within MNO organizations creates internal slack, which enables a flexible environment with dedicated resources and increased capacity for learning and exploration in unpredictable and diverse new technology markets (Bentley and Kehoe 2020). However, such gains from branded MVNOs in the 3G market diminish over time, suggesting the declining value of flexibility and internal slack for new technology exploration as the technology market matures. This alignment of value chain governance with exploitation (for efficiency) and exploration (for flexibility) is further supported by our additional findings. For example, the positive impact of third-party MVNOs on 2G market expansion is greater in markets with greater rural populations. This is because the access to rural populations is costly, highlighting the value of complementary resources provided by third-party MVNOs for efficient exploitation (GSMA 2016). On the other hand, the positive impact of branded MVNOs on 3G technology market expansion is greater in markets with greater young populations. The demand of young populations is more diverse, quickly shifting, and highly unpredictable, thus better suited for more flexible MNOs with internal slack.

This study makes the following contributions. First, we investigate value chain delegation of MVNOs from the theoretical lens of the exploitation and exploration

of technologies. The extant literature has studied exploitation and exploration in intrafirm settings or strategic alliances (Adler et al. 1999, Jansen et al. 2009, Lavie et al. 2011, Park et al. 2020). Our study focuses on value chain governance, particularly value chain delegation, in the context of overlapping generations of mobile network technologies, which extends the literature of exploitation and exploration to a value chain perspective. Our findings highlight the different effects of value chain governance on the trade-off between exploitation and exploration with respect to efficiency and flexibility, which makes an important contribution to the literature. Related literature has also studied technological transition from the theoretical lens of dominant design, which focuses on competition among new technology designs or between new and existing technologies (e.g., Anderson and Tushman 1990). However, our research extends the related literature by identifying and establishing the role of value chain governance in technology transitions. Second, overlapping technologies evolve as industry clockspeed transforms new technologies and renders existing technologies obsolete. Therefore, it is essential to manage value chains to adapt to such dynamics (Teece et al. 1997, Chandrasekaran et al. 2012, Posen and Levinthal 2012). By incorporating the dynamic development of overlapping technologies into value chain governance, we contribute to the literature by providing a dynamic perspective of the exploitation and exploration of fast-changing technologies in value chains. Third, our findings are not limited to 2G and 3G mobile technologies.² They provide broad insight for researchers, managers, and policymakers in value chain design with fast-evolving technologies. Future mobile technologies, such as the emerging 5G technology platforms, will enable network slicing and greater customization of the new generation of MVNOs (Grimaldo 2019). Managers need to develop more flexible and efficient value chain governance solutions for future technologies.

2. Mobile Network Technologies and MVNOs

Fast-changing technologies are associated with fast industry clockspeeds, high fixed costs, and high uncertainty, and so increasingly depend on flexible and efficient value chains (Christensen et al. 1998, Souza et al. 2004). In the case of the telecommunication industry, mobile network technologies present unique challenges:

- **Radical transformation.** In the history of mobile network technologies, 1G was an unmarred product without wide-spread commercialization, and 4G was largely an improvement in data transmission. However, the transformation from 2G to 3G radically changed mobile technology from voice-centric to data-centric services (Tilson and Lyytinen 2006, Ansari and Garud

2009). As a result, the launch of 3G created two independent markets with 2G and 3G technologies.

- **Slow diffusion.** The diffusion of 3G technology experienced an insufficient supply of compatible handsets, consumer hesitation toward the new technology, and other difficulties in its early stages (Fuentelsaz et al. 2008). After 3G had become available, 2G was still used to generate steady revenues and cash inflow for MNOs (Tilson and Lyytinen 2006), showing the MNOs' strategy to exploit the existing 2G networks for residual value.

- **Substantial cost.** Spectrum license auctions were widely introduced in the distribution of 2G licenses, leading to substantial license fees for MNOs (e.g., in the European Union, 100 billion U.S. dollars in total) (Tilson and Lyytinen 2006). Combining the cost to obtain 3G licenses with the construction cost of base stations, the unprecedented upfront costs of 3G exposed MNOs to great financial risks, further motivating MNOs to quickly explore the 3G market share.

- **Fast development.** In the case of the telecommunication industry, the development of mobile network technology has accelerated over time (Miranda and Lima 2013). 2G technology was introduced 12 years after 1G was introduced, but 3G technology came only 9 years after the launch of 2G. As a result, "telecommunication players are being forced to make high-stake multi-billion investments, even when the previous generation technologies have not generated desired returns" (Lincoln 2012, p. 179). The fast clockspeed of technology development has made it necessary for MNOs to continue to exploit existing technologies while exploring new technologies.

The radical disruption and slow diffusion of mobile network technologies have led to markets with overlapping generations of mobile technologies. Figure 2 shows the example of AT&T's technology transformation timeline. In particular, AT&T launched 2G services in 1996 and continued to offer them until 2016, overlapping with 3G starting in 2004. Managing such technologies poses a significant challenge for MNOs to quickly seize the new technology market while capitalizing on

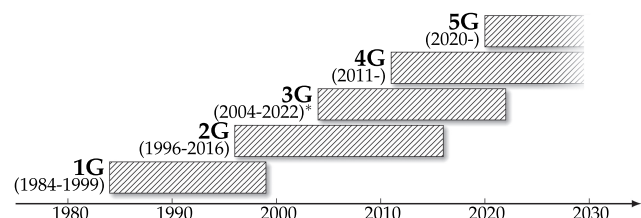
the remaining value from the market of existing technologies. To succeed, MNOs need to address two aspects of the overlapping 2G and 3G technologies: exploration in the 3G market and exploitation in the 2G market, based on the typology proposed by He and Wong (2004), in which **exploration** refers to activities aimed at entering a new product market, while **exploitation** refers to activities aimed at improving an existing product market.

Further, because of the substantial upfront costs, MNOs are constrained by limited financial and human resources in expanding the overlapped technology markets. MVNOs provide complementary assets (e.g., market know-how, sales skills, and distribution systems) for MNOs to address the issues of ambidexterity with overlapping generations of technologies (Jaspers et al. 2007). In doing so, MVNOs perform under the governance of the value chain, that is, the ownership of the MVNOs represented by branded and third-party MVNOs (Lehikoinen et al. 2014, Matin 2014). Without owning physical infrastructure or spectrum, the branded and third-party MVNOs complement their respective MNOs with assets of varying degrees of flexibility and efficiency, because the MNOs maintain ownership of the former and delegate ownership to the latter. The differences are shown in Figure 3 and, more formally, in the following definitions: **Branded MVNOs** are subsidiaries of MNOs and operate under the MNO brands (or are co-branded with the MNO). They are usually governed by a revenue-sharing contract. On the other hand, **Third-party MVNOs** are independent entities and obtain bulk service from MNOs through wholesale agreements.

The importance of MVNOs has further increased upon the arrival of 5G technology. Knect365 (2018) recently released an extensive industry report based on a survey collected from 500 telecommunication companies across the world. The results suggest that 69% of respondents named the implementation of new technologies, especially 5G, as the most critical factor that would reshape the industry and its supply chains. 5G not only offers faster speed, but it also allows network slicing and mini networks, which will lower market entry barriers and enable greater customizability for the new generation of MVNOs (Grimaldo 2019). For example, Sprint announced a partnership with a few MVNOs in the preparation stage of its 5G launches in select metropolitan areas in the United States (Fletcher 2019).

Lastly, managing value chains with overlapping generations of mobile technologies is increasingly a task of dynamic positioning rather than static alignment, given the industry clockspeed (Raisch et al. 2009). To respond to fast-developing technologies, firms need to assess and adjust their governance structure based on this dynamic environment (Teece et al. 1997, Westerman et al. 2006, Posen and Levinthal 2012). The fast-changing and coexisting nature of multiple generations of technologies

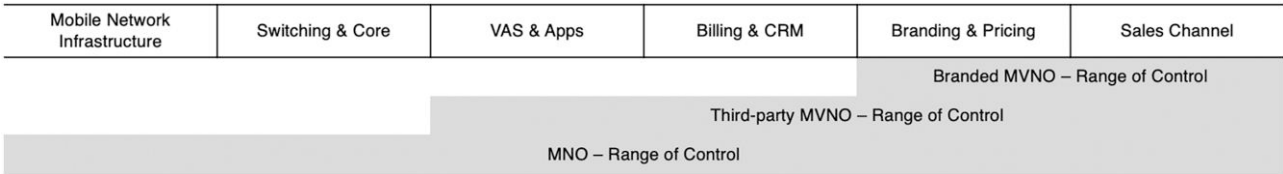
Figure 2. Timeline of AT&T's Mobile Network Technologies



Source: <https://investors.att.com/news-and-events/news-releases/>.
 Note. AT&T had divestiture, merger, and acquisition in the history, which is not denoted in the figure for simplicity.

*Though it completely phased out 3G in 2022, AT&T stopped taking new 3G customers in June 2019.

Figure 3. Ranges of Control for Branded MVNO, Third-party MVNO, and MNO



Note. VAS, value-added services; Apps, applications; CRM, customer relationship management.

further complicates value chain design and governance. This study, therefore, is highly timely and relevant because it provides causal evidence of the dynamic value chain design under overlapping generations of technologies.

3. Related Literature

Our study of value chain governance is conducted in the context of overlapping generations of mobile network technologies. Prior research has extensively studied the competition and substitution of technologies through the lens of dominant design theory (e.g., Soh 2010, Argyres et al. 2015). For multiple generations of technologies, dominant design theory primarily focuses on the competition within and between generations of technologies, where competing technologies often lead to the emergence of a dominant design, which captures a significant portion of the market share until the next technological disruption (Tushman and Anderson 1986, Anderson and Tushman 1990). Overlapping generations of mobile network technologies, however, involve well-recognized superior next-generation technologies that are developed and implemented collaboratively in the value chain of mobile networks (Leiponen 2008).

We theorize value chain delegation with overlapping generations of technologies as mechanisms of exploration and exploitation of technologies that coexist in value chains. Previous research has identified multiple mechanisms to achieve exploration and exploitation balance, which can be categorized by organizational structure: intrafirm and interfirm mechanisms (Raisch and Birkinshaw 2008, Park et al. 2020). Intrafirm mechanisms are the mainstream, including the following: structural differentiation, where organizations pursue both exploitation and exploration simultaneously (e.g., Siggelkow and Levinthal 2003, Jansen et al. 2009); temporal differentiation, where organizations focus on either exploitation or exploration sequentially and cycle back and forth (e.g., Adler et al. 1999, Liang et al. 2022); and contextual ambidexterity, where organizations create internal environments that promote the focus on exploitation and exploration respectively (e.g., Gibson and Birkinshaw 2004). On the other hand, the few interfirm studies focus on horizontal relationships such as strategic alliance (e.g., Rothaermel and Deeds 2004, Lin et al. 2007, Lavie et al. 2011), overlooking the vertical

structure, especially in the context of value chain management (Ramdan et al. 2021). Our paper complements this literature by investigating value chain delegation in the exploration and exploitation of technologies.

The telecommunication industry provides a perfect setting to study these questions: an MNO could adopt its own branded MVNO or delegate to a third-party MVNO, which we theorize as internal slack or complementary resources, respectively. Internal slack refers to the type of resource in a firm that is in excess of what is required for its current efficient operation (Mishina et al. 2004). Previous studies show that internal slack leads to increased adaptability and flexibility for market expansion (Nohria and Gulati 1996, Cheng and Kesner 1997, Voss et al. 2008, Bentley and Kehoe 2020). For example, Voss et al. (2008) find that internal slack is positively related to exploration and negatively related to exploitation in a high-risk external environment. Though internal slack fosters innovation, it may also be detrimental due to reduced discipline in resource management, leading to lax controls and inefficiency (Nohria and Gulati 1996). For example, Cheng and Kesner (1997) find that slack leads to reduced responsiveness to the external environment. However, in a dynamic environment, slack promotes exploration by creating a flexible environment for idea generation and innovation (Tan and Peng 2003, Mishina et al. 2004). Deb et al. (2017) find a positive relationship between slack and firm performance when the external environment requires greater adaptability, such as a highly competitive, high-tech, and growth-focused environment. Alternatively, delegation may allow firms to access complementary resources from other value chain members (Wu 2004), which supports greater reach into niche market segments and more efficient expansion (Lee and Whang 2002). By delegating, MNOs can quickly access niche market segments (e.g., industrial, demographic, and geographical) that are difficult or costly to reach, improving organizational efficiency (Varoutas et al. 2006, Jaspers et al. 2007). However, a third-party MVNO extends the value chain and complicates the processes of adjustment necessary for flexibility due to a decentralized decision-making process, particularly in an uncertain and dynamic environment (Brahm and Tarzījān 2016).

Our study is also related to industry clockspeed research in the literature on exploration and exploitation.

To survive and thrive in rapidly changing environments, firms need to continue to adjust current operations or create new initiatives to respond to uncertainty and stay competitive (Levinthal and March 1993, Jansen et al. 2006). Previous studies find that the importance of the balance between exploration and exploitation is more salient in fast-changing industries (Lee et al. 2015, Teece et al. 2016, Luger et al. 2018). Firms must continuously develop and manage internal and external resources to address fast changes (Sirmon et al. 2007, Teece et al. 2016). Specifically, Lee et al. (2015) find that exploration and exploitation improve a firm's agility to respond to uncertainty, particularly in dynamic environments. Similarly, Luger et al. (2018) find that the speed of change in an external environment is a contingent factor to exploration and exploitation for superior performance in a fast-changing external environment. Because resources are critical but constrained in balancing exploration and exploitation (March 1991, Kang and Kim 2020), firms must continuously develop and manage internal and external resources to address fast changes (Sirmon et al. 2007, Teece et al. 2016). In the context of overlapping generations of technologies, our research adopts a novel methodology and provides causal evidence of the dynamic patterns of exploration and exploitation in value chain governance.

4. Data and Exploratory Analysis

In this study, we explore the impact of MVNO launches on MNO market shares and how this impact varies across new and existing technologies. We collect data on MVNO launches and MNO market performance information from an independent research company specializing in the telecommunication industry. In this section, we first describe the proprietary data and then present a difference-in-differences (DID) model as an exploratory analysis.

4.1. Data Description

4.1.1. MVNO Launch Data. The data set of MVNO launches is acquired as a panel of 60 quarters from 2000Q1 to 2014Q4. The data contain information including names of MVNOs and their hosting MNOs (e.g., AT&T is the host of Cricket Wireless), launch dates of MVNOs, and the relationship between MNOs and MVNOs (branded or third-party). Our data cover 347 discount MVNO launches across multiple continents whose prices are lower than those of the MNOs. Discount MVNOs target markets for future growth where consumer willingness to pay is lower than that of MNOs (ITUNews 2019).

4.1.2. MNO Market Performance Data. From the same source, we obtain additional market performance information for MNOs, including total revenue, total

connections, 2G/3G market shares, and total market shares. This data set contains the quarterly market performance of MNOs from 178 countries ranging from 2000Q1 to 2014Q4. The MNOs in our sample are highly consistent with those on the list of the International Telecommunication Union (ITU). Figure 4 shows the distribution of MNOs and MVNO launches across different continents.

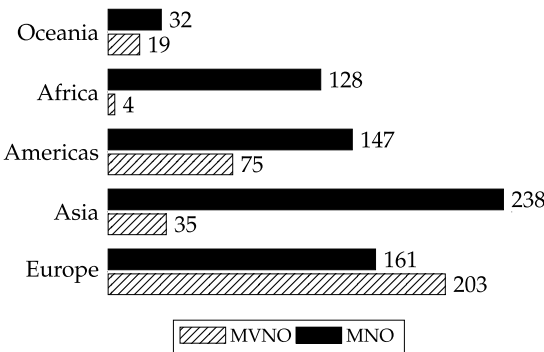
Prior to our analysis, we clean and merge the two described datasets. For the MNO performance data, we disqualify 34 MNOs in the overseas territories that have been listed as Non-Self-Governing Territories by the United Nations because overseas territories lack independence to regulate the telecommunication market and MNOs there are often managed by their counterpart in the governing countries.³ Second, we keep MNOs in countries that have at least three MNOs. Because we use 2G and 3G market shares as the performance measures of MNOs, MNOs in a country with too few operators tend to get extreme values of the market share. Third, we exclude data in Oceania because most MVNOs are launched by Australian MNOs while there are no other countries comparable in Oceania to serve in the control group. The final merged sample includes 603 MNOs and 306 MVNO launches covering 60 quarters (2000Q1 to 2014Q4).

4.2. Exploratory Analysis

In this section, we conduct a sliding-window-adjusted DID model as an exploratory analysis. The purpose of this step is to estimate the average impact of MVNO launches on MNO market shares.

We first explain the identification of treatment and control groups by introducing the key treatment indicators. We create two treatment indicators for the launches of branded (*Branded*) and third-party (*Thirdp*) MVNOs, respectively. *Branded* (*Thirdp*) is coded as 1 for all MNOs that launch branded (third-party) MVNOs and 0 for all MNOs that launch no MVNOs within our observed time range, regardless of the ownership. The posttreatment indicator (*After*) is coded as 1 for quarters after an MVNO launch and 0 for quarters before an

Figure 4. Total Number of MNOs and MVNOs by Continent



MVNO launch. Next, we introduce the dependent variables in the DID model. Since our data cover as many as 178 countries across the world, using an absolute measure of MNO market performance would bias our results because market sizes differ largely between countries. Thus, we employ two proportional measures, the market shares of 2G and 3G services (MS_{2g} and MS_{3g}), as our dependent variables to measure the performance of MNOs in the existing and new mobile network technology markets, respectively (Bayus and Putsis 1999). 2G and 3G market shares are calculated as:

$$MS_{2g_{it}} = \frac{Con_{2g_{it}}}{\sum_{i=1}^{n_i} Con_{2g_{it}}} \text{ and } MS_{3g_{it}} = \frac{Con_{3g_{it}}}{\sum_{i=1}^{n_i} Con_{3g_{it}}} \quad (1)$$

where $Con_{2g_{it}}$ and $Con_{3g_{it}}$ are the total number of subscribers on the 2G and 3G mobile networks of MNO i at quarter t . Given that MNO i operates in a country that has a total number of n_i MNOs, $\sum_{i=1}^{n_i} Con_{2g_{it}}$ and $\sum_{i=1}^{n_i} Con_{3g_{it}}$ represent the total number of 2G and 3G connections in the country. Using market shares as the dependent variable would be inappropriate if a country has a small number of MNOs because monopoly or oligarchy could result in the extreme value of market shares. For this reason, we drop the countries that have less than two MNOs. Following previous studies (e.g., Steenkamp and Geyskens 2014, Morais et al. 2016, Wang et al. 2016), we test alternative measures to correct the potential bias caused by the “sum-to-one” property of market shares (see Online Appendix B). The results remain consistent with our main results. The DID models we employ are:

$$MS_{it} = \beta_0 + \beta_1 After_t + \beta_2 Branded_i \times After_t + Z_{it}\Gamma + \lambda_i + \mu_t + \varepsilon_{it} \quad (2)$$

$$MS_{it} = \beta_0 + \beta_1 After_t + \beta_2 Thirdp_i \times After_t + Z_{it}\Gamma + \lambda_i + \mu_t + \varepsilon_{it} \quad (3)$$

where i and t denote MNOs and time periods, respectively. MS_{it} is a vector that includes 2G and 3G market shares of MNO i at quarter t . $Branded_i$ and $Thirdp_i$ are the binary variables that indicate if MNO i is treated. $After_t$ is the variable that indicates the posttreatment period. We capture the average treatment effect through β_2 , which is the estimated coefficient of $Branded_i \times After_t$ and $Thirdp_i \times After_t$. Z_{it} is a vector of the control variables, which includes average revenue per user (ARPU), net market share addition (MSA), total number of MVNOs launched (TV), quarters since the previous MVNO launch (GAP), gross domestic product per capita (GDPPC), and Herfindahl-Hirschman Index (HHI). We discuss the details of the control variables in Online Appendix A.3. Lastly, λ_i and μ_t are the MNO and quarter-year fixed effects, respectively. Because the MNO fixed effects subsume all time-invariant effects, the linear terms of the treatment group indicators ($Branded$ and $Thirdp$) are omitted from the equations.

Given that this analysis is for exploratory purposes, we keep the details about control variable construction, the matching of treated and control MNOs, the validation of parallel trends, and sliding window adjustment in Online Appendix A. We present the estimation results of Models (2) and (3) in Table 1. We find that launching branded MVNOs has a significant positive effect on the 3G market share ($\beta = 0.040, p < 0.01$) and launching third-party MVNOs has a positive impact on the 2G market share of the hosting MNOs ($\beta = 0.010, p < 0.01$). As predicted, branded MVNOs create a flexible environment through internal slack, which assists the hosting MNOs with accommodating the

Table 1. Exploratory Analysis: Estimation Results from the Sliding-Window-Adjusted Difference-in-Differences Models

Panel A: Branded MVNO			Panel B: Third-party MVNO		
D.V.	Model 1 MS_{2g}	Model 2 MS_{3g}	D.V.	Model 3 MS_{2g}	Model 4 MS_{3g}
<i>After</i>	0.000 (0.003)	−0.016 (0.012)	<i>After</i>	−0.008* (0.004)	−0.012 (0.010)
<i>Branded</i> × <i>After</i>	0.006 (0.004)	0.040** (0.013)	<i>Thirdp</i> × <i>After</i>	0.010** (0.004)	−0.007 (0.011)
Controls	Y	Y	Controls	Y	Y
MNO FE	Y	Y	MNO FE	Y	Y
Quarter-year FE	Y	Y	Quarter-year FE	Y	Y
Observations	366	211	Observations	220	115
R^2	0.480	0.798	R^2	0.454	0.511

Notes. All models include ARPU, MSA, TV, GAP, GDPPC, and HHI as control variables. The estimated coefficients of the control variables can be found in Table A.1 of Online Appendix A. Models 1 and 2 report the estimated coefficient values and robust standard errors (in parentheses) of Equation (2). Models 3 and 4 report the estimated coefficient values and robust standard errors (in parentheses) of Equation (3). Models using alternative standard error clustering methods yield consistent results (see Online Appendix H for details). We report within R^2 in the table. The bold font is used to highlight the key information.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$.

new technology and adapting to the new market (March 1991, O'Reilly and Tushman 2013). Moreover, efficiency is prioritized in the existing technology market. Delegating to a third-party MVNO provides the MNO direct access to complementary resources (e.g., strong representations of niche market segments), enabling the MNO to access underserved segments while remaining focused on its mass market (March 1991).

Notice the DID method adopted here only measures the average treatment effect across all MVNO launches in different years. However, as raised in our research question, environmental dynamism is rather important in the context of overlapping generations of technologies. Thus, we employ a more appropriate methodology that allows us to assess the impact of MVNO launches over time and therefore explore the dynamic aspects of the positive impact of branded MVNOs in the 3G market and that of third-party MVNOs in the 2G market.

5. Quasi-Experimental Design

To improve the rigor of the causal inference and, more importantly, identify trends in treatment effects as technology matures, we apply the method of dynamic propensity score matching (dPSM). In this section, we first introduce the method and explain why it is the preferred methodology given our research context. We then proceed to the construction and refinement of treatment and control groups. Finally, we elaborate on the matching procedure and measurement of treatment effects in our dPSM design.

5.1. dPSM – Empirical Strategy

The dPSM technique matches the treatment and control groups for each period to dynamically capture the treatment effects over time. Meanwhile, it retains the advantage of DID, which mimics a quasi-experimental design and creates two comparable groups with MVNO launches being the only difference (Rosenbaum and Rubin 1985).

Specifically, in each period, we construct two groups through dPSM: a treatment group of MNOs that launch only one MVNO in a specified time window, and a control group of MNOs that launch no MVNOs in the same time window. The pairing is based on their estimated propensity score, which represents the probability of launching MVNOs in each period. In this study, we construct one sample each for branded and third-party MVNOs, where the identification and matching of treatment and control groups are managed separately.⁴ We then, within each sample, compare the market share changes of the treated MNOs with the matched control MNOs in the posttreatment periods to assess the impact of the MVNO launches on the performance of the hosting MNOs. To explore the implications of MVNOs in

markets with different degrees of maturity, we also discuss the treatment effects of the branded and third-party samples in new and existing mobile technology markets.

The adoption of dPSM provides the following advantages: first, the traditional DID method that we use to explore the data allows us to evaluate the average treatment effect. However, we are unable to quantify treatment effects over time. With dPSM, we can measure treatment effects for each period and assess the changes in the trend of the treatment effects as technologies mature. Second, our sample size is relatively small because most countries have a limited number of MNOs and dPSM performs better in small samples (Rishika and Ramaprasad 2019). By using individual-level matching in each time period, dPSM grants a better matching outcome and thus a more rigorous evaluation of the treatment effects. dPSM has recently been adopted and validated by research studying similar problems (Aral et al. 2009; Aral and Van Alstyne 2011; Aral and Walker 2011, 2014; Bondonio and Greenbaum 2018; Rishika and Ramaprasad 2019), and has been shown to be effective in constructing causal inferences in observational datasets.

5.2. dPSM – Matching Algorithm

We first discuss the identification and refinement of the treatment group and the candidate control group. We then show the details of the matching procedure, followed by an example to demonstrate the algorithm. In the rest of this section, we first select the parameters in the estimation process. With the application of the parameters, we then discuss the detailed dynamic matching process. Finally, we conduct quality matching checks in the matched sample.

1. Identification of the treatment group. To identify the treatment group, we use the following steps. We begin with a cleaned sample as described in Section 4.1, which contains 306 MNOs that launched at least one MVNO during the time period covered in this study. To maintain a consistent treatment level, we focus on MNOs that launch only one MVNO in one period (a quarter-year). We thus exclude 126 MNOs with multiple MVNO launches in the same period, which leaves us with 50 MNOs in the treatment group of the branded sample and 130 MNOs in the treatment group of the third-party sample. We then further apply the restriction that treated MNOs cannot launch more than one MVNO in k_t neighboring quarters. This is because if an MNO launches a second MVNO in the posttreatment periods of the previous launch, it would be impossible to completely disentangle the treatment effects of the two launches. Specifically, we set $k_t = 6$ because we later use the average covariate value of 4 quarters after the launch to measure the treatment effect. If k_t is set to be equal to or less than 4, the posttreatment period of an MVNO launch could overlap with the pretreatment

period of another launch. Such overlaps would contaminate the causal inference. For this reason, we include 2 extra quarters as a buffer to further ensure that all treatment effects of MVNO launches are estimated independently from each other. With this restriction, 91 MVNO launches are further removed. The refined treatment group contains 33 MNOs in the branded sample and 56 in the third-party sample.⁵

2. Identification of the candidate control group. To identify the candidate control group for each treated MNO, we use the following steps. For the candidate control group of a treated MNO, we first identify the MNOs that are in another country on the same continent. This helps mitigate the zero-sum situation created by the use of market shares as dependent variables⁶ and improve the matching quality by ensuring a sufficiently large candidate pool for the matching algorithm.⁷ Second, countries in the same continent are expected to share similar characteristics, such as market competition, government regulations, and consumer behavior. Thus, we apply the restriction that treated MNOs can only be matched with control MNOs in the same continent to minimize the differences between the two groups.

Next, we proceed to refine the candidate controls by removing MNOs that launch MVNOs in the k_c periods neighboring a treated MNO's launch period t . If an MNO in the candidate control group is treated in the pre- or posttreatment period, it would be inappropriate to serve as a benchmark in the measurement of the treatment effects. This refinement step ensures that MNOs in the candidate control group are untreated. Similar to k_t , we set $k_c = 6$ and remove the MNOs from the candidate control group if they have an MVNO launch in the neighboring 6 periods of the given launch period, regardless of the governance of the MVNO in the launch. This process is conducted for each launch period to ensure that the MNOs in the candidate control group are suitable to be matched with the corresponding treated MNOs. For example, in Figure 5, given that the MVNO is in Quarter 12, MNOs that have MVNO launch records between Quarter 6 and Quarter 18 will be disqualified from the candidate control group. We also exclude the MNOs operating in a market that has other MNO launches, closures, or mergers. Because we use market share as the dependent variable, the changes in the total number of MNOs in the market may cause unintended surges or drops in the market share, which could contaminate the measure of the treatment effects. Thus, we further remove all MNOs operating in a market that has inconsistent total numbers of MNOs within the pre- and posttreatment periods (from Quarter $t - k_p + 1$ to Quarter $t + k_c$).

After applying the above matching restrictions, we construct two samples for branded and third-party MVNOs. In the branded sample, the treatment group

consists of 33 MNOs with MVNO launches. In the third-party sample, the treatment group consists of 56 MNOs.⁸ The candidate control groups of the two samples are constructed for each period based on the corresponding treated MNOs in the same periods. The candidate control group identified is in proportional size to the treatment group and thus ensure appropriate dynamic matching in the next step (see Online Appendix C.1 for details).

3. Dynamic propensity score matching. To model the probability of an MVNO launch, we first select matching variables for the estimation of the propensity score. We select several matching covariates at the MNO and country level to represent different characteristics of MNOs in the pretreatment periods, including total connections (Con_tot), total market share (MS_tot), 2G market share (MS_2g), total market penetration (MP_tot), and GDP (GDP).⁹

- Total Connections (Con_tot) is the total number of subscriber identification module (SIM) cards registered on the mobile network, which is a proxy for the size of an MNO. MNOs with similar numbers of subscribers are usually similar in size and are more likely to adopt the same strategies in market expansion, such as launching MVNOs.

- Total Market Share (MS_tot) is the ratio of connections of an MNO to the total connections of the market, which is a proxy for an MNO's general competitiveness in the market. Specifically, a higher market share represents greater competitiveness and higher market power relative to other MNOs in the same market. Thus, MNOs with similar market shares will likely adopt similar expansion strategies with similar probabilities to launch MVNOs. The total market share of MNO i at time period t is calculated as $MS_tot_{it} = (Con_tot_{it}) / (\sum_{i=1}^{n_i} Con_tot_{it})$, where Con_tot_{it} is the total number of SIM cards registered on the mobile network of MNO i at time t and n_i is the number of MNOs in the market that contains MNO i . Thus, $\sum_{i=1}^{n_i} Con_tot_{it}$ represents the total number of subscribers registered to all n_i MNOs.

- 2G Market Share (MS_2g) is the ratio of an MNO's 2G connections to the total 2G market connections. In our data period, 2G was the mainstream market, where the majority of MNOs had already adopted 2G for years prior to the beginning of our observed period. Though some MNOs launched 3G, they were replacing 2G only slowly. Thus, we use 2G market share as a proxy for a firm's core competitiveness, which captures MNO competitiveness in the major market.

- Total Market Penetration (MP_tot) is the ratio of an MNO's subscribers to the total population of the country, which captures not only the size of the MNO but also the size of the market. The total

market share of MNO i at time t is calculated as $MP_{tot_{it}} = Con_{tot_{it}} / Pop_{it}$, where $Con_{tot_{it}}$ is the total number of SIM cards registered on the mobile network of MNO i at time t and Pop_{it} is the total population of the Country j that MNO i operates in.

- Gross Domestic Product (GDP) is the logarithm-transformed gross domestic product per capita, which captures the country's economic development and stability in which an MNO operates. A higher GDP usually represents a more developed economy with a greater market size and higher purchasing power. Thus, MNOs may consider the GDP of the market as a factor in MVNOs launches.

After matching covariates are carefully selected, we proceed to the quarter-by-quarter estimation of the propensity scores. Because an MNO could decide to launch an MVNO at any time period, we treat all launch periods as discrete and estimate the propensity scores for each quarter that has an MVNO launch (Sianesi 2004). The propensity score value represents the probability that an MNO would launch a branded or third-party MVNO, which is estimated from a logistic regression model of the matching covariates. Specifically, we identify all launch periods that have MVNO launch records in our time range and use the average value of matching covariates in the previous k_p periods as the benchmark to estimate the propensity score. For example, if an MVNO is launched in Quarter t , we consider the value of matching covariates from Quarter $t - k_p + 1$ to Quarter t and estimate logistic regression on the average value of the interested quarters to estimate the propensity score.

MNOs may consider recent performance in the MVNO launching decision, and therefore we set $k_p = 4$ and consider the average value of matching covariates in the previous 4 periods as the benchmark in the propensity score estimation for a given period.¹⁰ Because dynamic propensity score matching requires the construction of treatment and control groups for each time period, we estimate the logistic regression for each period independently if any MNO launches an MVNO in the given period. The logistic regression model used to estimate the propensity score of MNO i 's MVNO launch at Quarter t is given by:¹¹

$$Pr\left(Treated_{it} = 1 \middle| \frac{1}{4} \sum_{k=t-4}^{t-1} X_{ik}\right) = \Phi\left(\sum_{k=t-4}^{t-1} X_{ik}\beta\right) \quad (4)$$

where $Treated_i$ is the treatment indicator of MNO i , which is $Branded_i$ in the branded sample and $Thirdp_i$ in the third-party sample. X_{ik} is a vector of covariates of MNO i at Quarter k , including total market share (MS_{tot}), 2G market share (MS_{2g}), total number of connections (Con_{tot}), total market penetration (MP_{tot}),

and gross domestic product (GDP) of MNO i at Quarter k , and $\sum_{k=t-4}^t X_{ik}$ is a vector of average values of the matching covariates in all of the previous periods (from Quarter $t - 4$ to Quarter t).

Once we obtain the estimated propensity scores, we use nearest neighbor matching (NN1) to construct the control group with the tolerance level of 0.05. Specifically, for each launch period, we pair treated MNOs with MNOs in the candidate control group that have the closest propensity score (Guo and Fraser 2015). After matching, our final sample for branded MVNOs includes 33 treated MNOs with 33 paired control MNOs. The final sample for third-party MVNOs includes 56 treated MNOs and 56 paired control MNOs. Our final sample covers MNOs across continents and both developed and developing countries. To ensure matching quality, we further check for substantial differences between treatment and control groups by testing the covariate balance in the pretreatment period. We test the differences between the two groups by propensity score plots and the matching bias estimation method proposed by Rosenbaum and Rubin (1985). The results suggest that our matching algorithm significantly reduces the bias between treatment and control groups (see Online Appendix C.2). Further, we show that the treated and matched control MNOs are comparable in the sense that they are close in all characteristics except MVNO launch (see Online Appendices D.1 and D.2) (Aral et al. 2009, Rishika and Ramaprasad 2019).

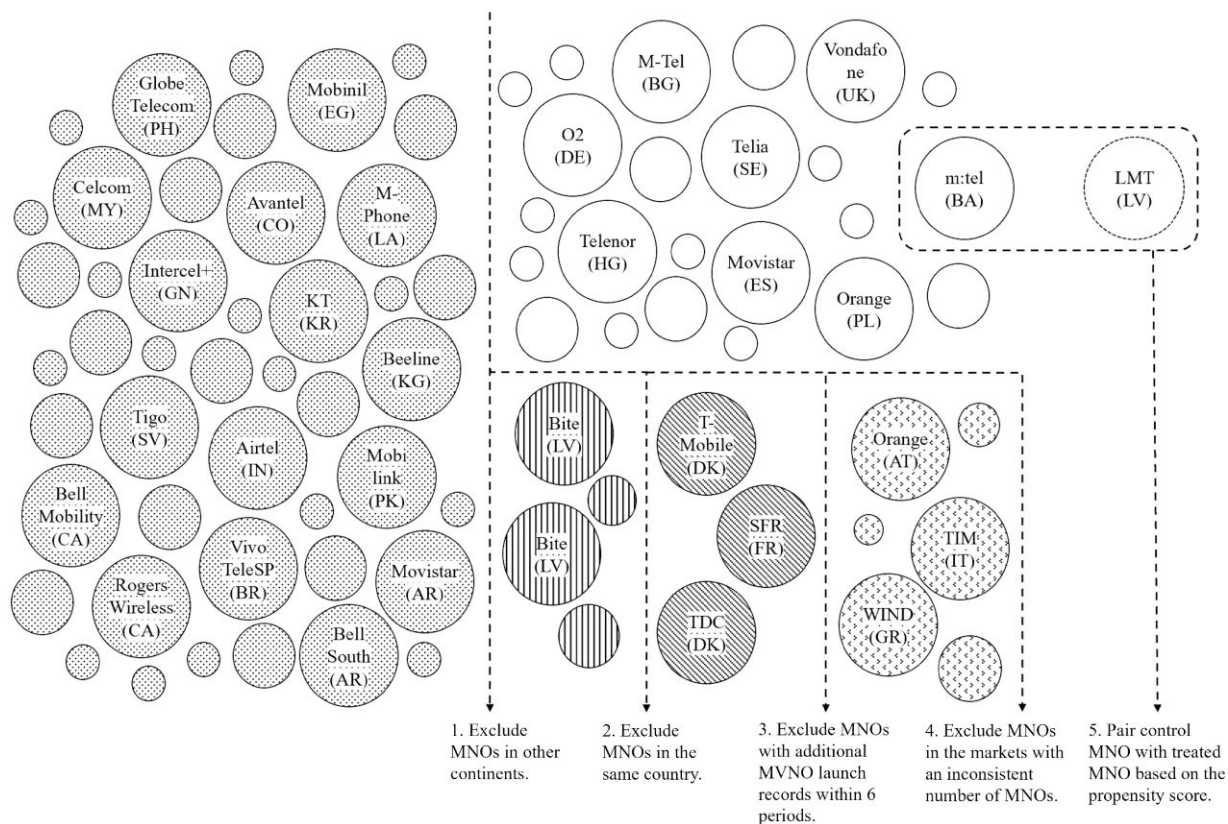
To help better explain the construction procedure of the treatment and control groups, we summarize the matching algorithm in Figure 6.

Further, in Figure 5, we demonstrate the matching algorithm by showing a step-by-step example of pairing a treated MNO with a comparable MNO in the candidate control group.¹²

5.3. dPSM – Measuring Treatment Effects

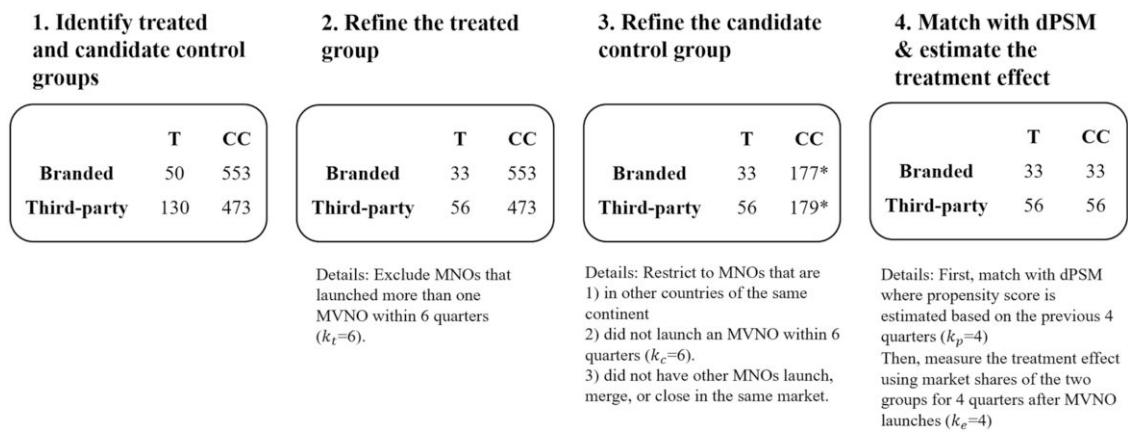
To estimate the treatment effect, we use the 2G and 3G market shares as performance measures of MNOs in existing and new technology markets, respectively. The treatment effect is measured by the differences in 2G (3G) market shares between the treatment and control groups in k_p quarters before and k_e quarters after launch, where k_e is the parameter of posttreatment period length. For each quarter that has an MVNO launch, we calculate the first difference between pre- and posttreatment periods for each group. Then, we group MVNO launches based on their launch periods and use the t -test to compare the second difference between the two groups year by year. We conduct the comparison on a yearly basis because presenting the treatment effect in an aggregate fashion would not capture seasonal effects. It therefore projects a clear trend for the impact of MVNO launches over time. The t -test

Figure 5. Example of Matching a Pair of Control and Treated MNOs



Notes. The text in the circle denotes each MNO's name and the country abbreviation in parentheses. Thus, in the figure, an MNO named LMT in Latvia launched Orate, an MVNO, at Quarter 12 (2003Q1). To find a comparable control MNO for LMT, we first exclude all MNOs outside of Europe, as well as the MNOs in Latvia. Setting $k_c = 6$, we identify three MNOs that have MVNO launch records from Quarter 6 to Quarter 18 and drop them from the candidate control group. We then exclude MNOs in France, Italy, and Denmark due to inconsistent numbers of MNOs in the market. Finally, for LMT and the remaining MNOs in the candidate control group, we estimate the propensity score based on the average value of the covariates from Quarter 8 to Quarter 11 ($k_p = 4$). Based on the proximity of estimated propensity scores, we pair an MNO named m:tel in Bosnia and Herzegovina as the counterfactual to LMT. To measure the treatment effect, we compare the mean difference between the market shares of the two MNOs.

Figure 6. Process of the dPSM Design



Note. In the boxes, T denotes the treatment group, CC denotes the candidate control group before matching, and C denotes the matched control group based on dynamic propensity score matching.

*Because the candidate control group is refined for each launch period, the number shown here is the illustration of MNOs launching MVNOs in the first period.

statistic is calculated as:

$$t = \frac{\bar{m}_t - \bar{m}_c}{\sqrt{s^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad (5)$$

where $\bar{m}_t$ and $\bar{m}_c$ are the sample mean for the treatment and control groups, respectively, which are calculated as follows:

$$\bar{m} = \frac{1}{k_e} \sum_{i=1}^{n_\tau} \left(\sum_{t=\tau+1}^{\tau+k_e} MS_{it} - \sum_{t=\tau-k_p+1}^{\tau} MS_{it} \right) \quad (6)$$

where MS_{it} is the 2G or 3G market share ($MS_{2g_{it}}$ or $MS_{3g_{it}}$) of MNO i at year t . n_τ represents the total number of MNOs for the year of launch period τ . The sample mean of treatment group ($\bar{m}_t$) is measured by the difference between the average market share of MNO i in pretreatment periods $[\tau - k_e + 1, \tau]$ and posttreatment periods $[\tau + 1, \tau + k_e]$. The corresponding sample mean of control group ($\bar{m}_c$) is calculated using the same method. We then run the t -test on the sample between the two groups to quantify the treatment effect. The 95% confidence interval is constructed by bootstrapping with 1,000 iterations (Austin and Small 2014).

Specifically, we set $k_e = 4$ and utilize the market share of 4 quarters after the MVNO launches to assess the treatment effect.¹³ We believe this is the appropriate setting to observe the impact of an MVNO launch for two reasons. First, the niche market segments targeted by MVNOs are usually smaller in size, which allows faster gains in market share. Second, following an MVNO launch, competitors in the same market may follow suit or take countermeasures, which could contaminate causal evaluation if the posttreatment period k_e is set longer. Additionally, to account for MNO- and market-level heterogeneity, we further include the same set of

control variables Z_{it} as that in the exploratory analysis of Section 4.2. The details of the control variables are included in Online Appendix A.3. Next, we discuss the estimated treatment effects in 2G and 3G markets and connect these results to those from the DID models.

6. Results

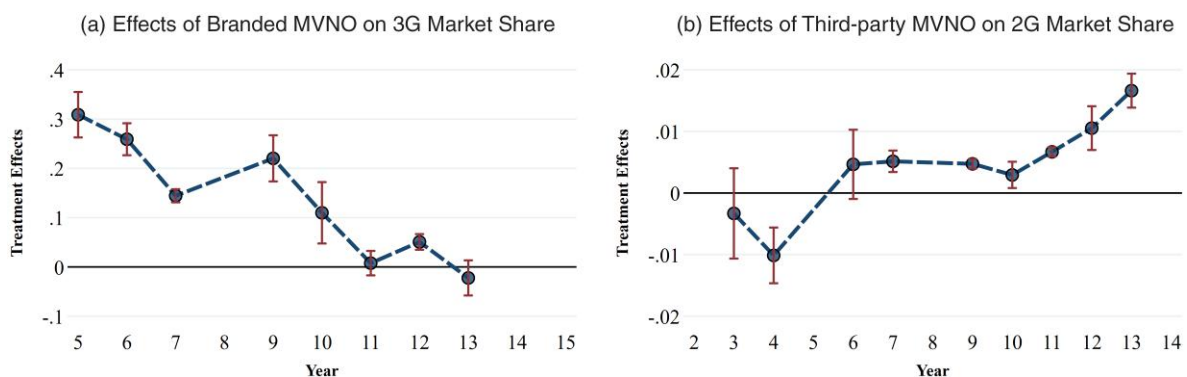
In this section, we discuss the dynamic propensity score matching (dPSM) results. We present estimated treatment effects of dPSM and further validate our results with robustness tests.

6.1. Treatment Effects

To assess the treatment effect, we compare the 2G and 3G market shares between the treated and matched control groups. Following the direction shown by the DID models in the exploratory analysis, we focus on the impact of branded MVNOs in the 3G market and that of third-party MVNOs in the 2G market. In Figure 7, panel (a) shows the estimated treatment effects of branded MVNO launches on the 3G market shares of the hosting MNO. Similarly, panel (b) shows the estimated treatment effects of third-party MVNO launches on the 2G market shares of the hosting MNO. In each panel, the vertical axis represents the estimated treatment effects, and the horizontal axis represents the calendar year, which we use as proxy for technology maturity.¹⁴ We next interpret and discuss the impact of branded MVNOs in the 3G market and third-party MVNOs in the 2G market respectively, through the corresponding panels in Figure 7.

6.1.1. Third-party MVNOs in the 2G Market. Figure 7(b) shows the treatment effect of launching third-party MVNOs on the 2G market shares of the hosting

Figure 7. (Color online) Treatment Effects of Branded MVNO Launch and Third-party MVNO Launch after Dynamic Propensity Score Matching (2G/3G Market Shares of Treatment Group – 2G/3G Market Shares of Control Group)



Notes. (a) The estimated treatment effects of branded MVNO launches on the 3G market share of the hosting MNOs. This figure shows the mean difference in the 3G market share between the treatment group (MNOs who launch branded MVNOs) and the matched control group. (b) The estimated treatment effects of third-party MVNO launches on the 2G market share of the hosting MNOs. This figure shows the mean difference in the 2G market share between the treatment group (MNOs who launch third-party MVNOs) and the matched control group. To construct the 95% confidence intervals, we use bootstrapping with 1,000 replications.

MNOs. We find that, over the years, the average difference in 2G market shares between MNOs that launch third-party MVNOs and the matched counterfactual control MNOs is 1.53% ($p < 0.001$). Though the first three years are significantly negative or insignificant, the treatment effects of third-party MVNO launches are significantly greater than zero and show an upward-moving pattern over the years. The increasing trend of third-party MVNOs is statistically significant in the slope test (see Online Appendix F for details). This indicates that, as existing technology markets mature, efficiency in market expansion becomes increasingly important (March 1991), highlighting the importance of complementary resources from delegation (Lee and Whang 2002, Wu 2004). In short, consistent with the exploratory analysis, we find that launching third-party MVNOs increases the 2G market shares of hosting MNOs. Assisted by the method of dynamic propensity score matching, we further discover that the positive effect of third-party MVNO launches has a clear increasing pattern as the technology market matures.

6.1.2. Branded MVNOs in the 3G Market. Figure 7(a) shows the treatment effect of branded MVNO launches on the 3G market shares of the hosting MNOs. The average yearly difference in 3G market shares between treated and matched control MNOs is 13.71% ($p < 0.05$). Additionally, the estimated treatment effects for each year are mostly significantly greater than zero, and we also observe a decreasing trend of such positive treatment effects as the market matures over time. The decreasing trend is significant at 95% in the slope test (see Online Appendix F for details). This indicates that launching branded MVNOs creates internal slack, which promotes flexibility and assists MNOs' expansion into new market segments (Voss et al. 2008, Bentley and Kehoe 2020). As the technology matures, however, the need for efficiency prevails over the need for flexibility (March 1991), which exposes branded MVNOs' disadvantage of inefficiency in utilizing resources (Nohria and Gulati 1996, Cheng and Kesner 1997).

To summarize, in this section, our main findings advance the average treatment effects by showing the variation of the treatment effects between branded and third-party MVNOs as technologies mature. Specifically, dPSM reinforces the average treatment effects: launching third-party MVNOs increases the market share of the hosting MNOs in the mature 2G technology market, whereas launching branded MVNOs increases the 3G market of the hosting MNOs. More importantly, dPSM provides dynamic patterns in the impact of MVNO launches under the two governance structures, where we find asymmetric patterns in the positive impacts between branded and third-party MVNOs: in the 2G market, the positive impact of third-party MVNO launches increases over time as the market matures; in

the 3G market, the positive impact of branded MVNO launches diminishes as the market matures over time.

6.2. Robustness Tests

We conduct several additional analyses to ensure that our estimated treatment effects from the dPSM analysis are robust. First, we test them using the Stable Unit Treatment Values Assumption (SUTVA), including covariate balance, parallel trends, and reverse causality. Second, we redefine the treatment and candidate control groups by setting more rigorous selection windows (k_t and k_c). Third, to validate the robustness of our matching process, we test an alternative set of matching variables. Additionally, we set a different matching window (k_p) and specify different numbers using nearest neighbor matching. Fourth, to test the validity of the estimation of the treatment effects, we apply a different posttreatment window (k_e). Lastly, we test the robustness of our findings on the transition between 3G and 4G.

6.2.1. SUTVA. To ensure matching quality, we first check the covariate balance between the treatment group and the matched control group. The results suggest that there is no statistically significant difference between the two groups (see Online Appendix D.1). Additionally, following Autor (2003), Dhanorkar (2019), and Chu et al. (2021), we test the parallel trends assumption by estimating the treatment effects by period. The estimated coefficient of the “leads” (i.e., quarters in the pretreatment periods) is insignificant, which suggests that there is no significant difference between the treatment and control groups identified in each quarter of the pretreatment periods (see Online Appendix D.2). Lastly, we test the Cox proportional hazard models and lag models (Dhanorkar 2019), from which we find no evidence of reverse causality (see Online Appendix D.3 for details).

6.2.2. Identification Strategy. To validate the construction of the treatment and control groups, we apply an alternative set of selection window parameters to redefine the treatment and control groups. In our main analysis, if the gap between two MVNO launches is less than six periods, we exclude both launches from the treatment group ($k_t = 6$). We now apply a more rigorous condition by setting a longer gap window ($k_t = 8$), where we are able to better avoid violating SUTVA. The results are highly consistent with our main results (see Figure E.1 of the Online Appendix). Furthermore, for the construction of the candidate control group, we similarly apply a more rigorous condition by setting a longer gap window ($k_c = 8$). Again, the results show high consistency with our main results (see Figure E.1 of the Online Appendix).

6.2.3. Matching and Pairing. Next, we check the robustness of the propensity score matching process. We first

use an alternative set of matching covariates. Specifically, we use net market share addition (*MSA*) as an alternative to total market share to capture the competency and market power of MNOs and inflation rate as an alternative to GDP to capture the economic development of a country. The results after applying the alternative matching variables remain qualitatively consistent (see Figure E.2 of the Online Appendix). Second, in our main analysis, we use the average value of all matching variables of 4 previous periods to estimate the propensity score in each period. However, though recent performance is likely to weigh more in an MNO's decision to launch MVNOs, it is possible that MNOs could consider their performance further back. Thus, we test an alternative dPSM algorithm that uses an extended window to construct matching covariates ($k_p = 6$). We find that our main results remain qualitatively consistent with this longer window of matching covariates (see Figure E.2 of the Online Appendix). Further, we use one nearest neighbor matching (NN1) in our main analysis. In the matching process, we match the treated MNOs with control MNOs that have the closest propensity score at each quarter. We now also test two and three nearest neighbor matching (NN2 and NN3) and find that our main results are still robust (see Figure E.3 of the Online Appendix). Lastly, to increase the power of the analyses, we perform optimal full matching as the alternative matching method. Optimal full matching assigns weights to all observations and thus potentially utilizes more information in the raw data set. Even under an extreme full matching, we find that the insights are still robust (see Figure E.4 of the Online Appendix).

6.2.4. Measurement of Treatment Effect. After MNOs in the treatment group are paired with comparable control MNOs, we use the 4 following quarters after a launch as the posttreatment periods and conduct *t*-tests to compare the difference in the 2G and 3G market shares between the two groups. Though we find positive treatment effects, it is also possible that the MVNO launch may have a different impact on the MNO in the long run. Thus, we apply a shorter and longer window to measure the performance change after the MVNO launch ($k_e = 2$ and $k_e = 8$) and find that our main results remain consistent (see Figure E.5 of the Online Appendix).

6.2.5. Generalizability Beyond 3G. One caveat in our analysis is that it is technology-specific, leaving open the possibility of the findings being limited between 2G and 3G technologies. To improve generalizability, we further examine delegation during the transition between 3G and 4G technologies. We replicate our analysis using additional data of 4G technology and our main results hold for 3G and 4G technologies (see Appendix E.4 for details). Specifically, we find that

launching branded MVNOs increases the market share of the new technology, that is, 4G, while launching third-party MVNOs increases the market share of the existing technology, that is, 3G, which shows that our findings are robust and generalizable for 3G and 4G technologies.¹⁵

7. Complementary Resource or Internal Slack

We have shown that MVNOs can be adopted as internal slack or complementary assets for MNOs to effectively access and expand to different technology markets. These markets may be presented by specific geographic and demographic groups with different technology preferences and market access costs. Specifically, the rural population is a market segment that is costly for MNOs to reach, and we argue that achieving high efficiency from complementary resources is critical. We thus examine whether the treatment effects of launching MVNOs with 2G technology vary based on the proportion of the rural population. Similarly, the young population is a market segment with a strong appetite and diverse need for new technologies. We argue that creating internal slack for flexibility should be more effective to appeal to this population. We thus examine whether the treatment effects of MVNO launches with 3G technology vary based on the proportion of the young population.

To conduct these tests, we collect rural and young population information from the report published by the World Bank.¹⁶ We measure rural population (*RP*) as the proportion of the rural population of a country in a given quarter. The measure for young population (*YP*) is constructed similarly. Following Ahmad et al. (2019), we define the mobile market segment of the young population as those aged 20–40. Mobile users in this age range are more interested in the new development of mobile technologies and thus more likely to become early adopters of 3G with more diverse 3G service plans.

Following Jola-Sanchez and Serpa (2021), we add the two population proportion measures as moderators in the DID models to explore how the two factors moderate the relationship between MVNO launches and MNO performance. First, we add linear and interaction terms for rural population to the DID model in Equation (3) and test the impact of a third-party MVNO launch on the 2G technology market share. Similarly, we add linear and interaction terms for young population to the DID model in Equation (2) and test the impact of a branded MVNO launch on the 3G technology market share. The two models are specified as follows:

$$\begin{aligned}
 MS_{it} = & \beta_0 + \beta_1 After_t + \beta_2 YP_{it} + \beta_3 Branded_i \times After_t \\
 & + \beta_4 Branded_i \times YP_{it} + \beta_5 Branded_i \times After_t \\
 & \times YP_{it} + \beta_6 RP_{it} + \mathbf{Z}_{it}\Gamma + \lambda_i + \mu_t + \varepsilon_{it} \quad (7)
 \end{aligned}$$

$$\begin{aligned}
MS_{it} = & \beta_0 + \beta_1 After_t + \beta_2 RP_{it} + \beta_3 Thirdp_i \times After_t \\
& + \beta_4 Thirdp_i \times RP_{it} + \beta_5 Thirdp_i \times After_t \times RP_{it} \\
& + \beta_6 YP_{it} + Z_{it}\Gamma + \lambda_i + \mu_t + \varepsilon_{it}
\end{aligned} \quad (8)$$

where *Branded* and *Thirdp* are the treatment indicators for branded or third-party samples. Z_{it} , λ_i , and μ_t are defined the same as those in the main analysis. We present the estimation results in Table 2. In Table 2 Model 2, we find that the three-way interaction between the treatment indicator, posttreatment indicator, and young population ($Treated_i \times After_t \times YP_{it}$) is positive and significant ($\beta = 1.496, p < 0.001$), suggesting that the impact of launching branded MVNOs on the 3G market share of the corresponding MNOs is greater in countries with a higher proportion of the young population. This finding supports the argument that branded MVNOs, as internal slack, allow more flexibility and adaptability in the exploration of the 3G market. The demand for mobile technologies in the young population is more diverse, quickly shifting, and highly unpredictable, thus better suited for more flexible MNOs.

In Table 2 Model 3, we find that the three-way interaction between the treatment indicator, posttreatment indicator, and rural population ratio ($Treated_i \times After_t \times RP_{it}$) is positive and significant ($\beta = 0.036, p < 0.01$), suggesting that the impact of launching third-party MVNOs on the 2G market share of the corresponding MNOs is greater in countries with a higher proportion of the rural population. This finding supports our

argument that delegation is more efficient and important in the exploitation of the 2G market because access to the rural population costs twice as much as it does to urban areas (GSMA 2016).

8. Discussion and Conclusion

In this study, we explore how introducing MVNOs impacts the hosting MNOs' market performance with overlapping generations of technologies. Our findings indicate that, with branded MVNOs, MNOs gain flexibility by creating internal slack, which is critical for exploring new technology markets, that is, 3G. On the other hand, delegating to third-party MVNOs leverages complementary resources to increase efficiency in the value chain for exploiting existing technology markets, that is, 2G.

We further show that the impact of MVNO launches under these two governance structures is contingent on the dynamics of the technology markets. For existing technologies, that is, 2G, launching a third-party MVNO would increase the market share of the hosting MNO in an increasing pattern over time. Delegating to a third-party MVNO allows the hosting MNO to take advantage of the complementary resources of the MVNO, which is increasingly beneficial in a mature technology market where the remaining market segments are more costly to reach (Jaspers et al. 2007). For new technologies, that is, 3G, launching a branded MVNO increases the 3G market share of the hosting

Table 2. Extension: The Estimation Results of Rural Population and Young Population

Panel A: Young Population and Branded MVNO			Panel B: Rural Population and Third-party MVNO		
D.V.	Model 1 <i>MS_2g</i>	Model 2 <i>MS_3g</i>	D.V.	Model 3 <i>MS_2g</i>	Model 4 <i>MS_3g</i>
<i>After</i>	0.003 (0.003)	−0.045* (0.019)	<i>After</i>	−0.011** (0.004)	−0.001 (0.010)
<i>YP</i>	−0.213 (0.387)	−0.430 (8.121)	<i>YP</i>	−0.381 (0.269)	−1.969 (1.371)
<i>RP</i>	0.399 (0.644)	−3.488 (4.158)	<i>RP</i>	−0.763 (0.718)	−1.186 (1.584)
<i>Branded</i> × <i>After</i>	0.020 (0.087)	−0.933*** (0.167)	<i>Thirdp</i> × <i>After</i>	0.016** (0.005)	−1.186 (0.035)
<i>Branded</i> × <i>YP</i>	−0.028† (0.021)	2.600 (8.474)	<i>Thirdp</i> × <i>RP</i>	1.135 (1.047)	−6.762 (6.445)
<i>Branded</i> × <i>After</i> × <i>YP</i>	−0.015 (0.128)	1.496*** (0.254)	<i>Thirdp</i> × <i>After</i> × <i>RP</i>	0.036** (0.012)	0.127 (0.083)
Controls	Y	Y	Controls	Y	Y
MNO FE	Y	Y	MNO FE	Y	Y
Quarter-year FE	Y	Y	Quarter-year FE	Y	Y
Observations	359	209	Observations	193	106
<i>R</i> ²	0.568	0.387	<i>R</i> ²	0.746	0.710

Notes. Models 1 and 2 report the estimated coefficient values and robust standard errors (in parentheses) of Equation (7). Models 3 and 4 report the estimated coefficient values and robust standard errors (in parentheses) of Equation (8). Note the linear terms of treatment group indicators (*Branded* and *Thirdp*) are omitted due to the MNO-level fixed effects. All models include the same set of control variables as that in Table 1 except use 2G HHI to accommodate rural population. We report within R^2 in the table. The bold font is used to highlight the key information.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$.

MNO in a decreasing pattern over time. A branded MVNO creates internal slack, which increases flexibility for MNOs to meet the unpredictable demand in the new technology market (Voss et al. 2008, Bentley and Kehoe 2020). This branded MVNO advantage, however, diminishes as the demand in the new technology market stabilizes over time.

8.1. Implications for Theory

The extant literature has extensively studied the importance of organizations pursuing exploration and exploitation in managing technologies (Levinthal and March 1993, O'Reilly and Tushman 2004, Chandrasekaran et al. 2012). The exploration and exploitation literature largely focuses on the intrafirm aspects of exploitation and exploration (e.g., Jansen et al. 2009), with the exception of strategic alliance as an area for interfirm studies (e.g., Lavie et al. 2011). We contribute to this literature by studying a vertical relationship, that is, delegation, with overlapping generations of technologies from the theoretical lens of exploitation and exploration. Further, the impact of delegation evolves quickly with the fast clockspeed of technology development, such as overlapping generations of mobile network technologies, and yet existing literature has seen limited analyses highlighting the importance of such dynamics in value chain governance or exploitation and exploration. Using a novel method that is highly relevant to studying such dynamic issues, that is, dPSM, we investigate the temporal impact of delegation with overlapping generations of technologies, which provides a new and dynamic view of the exploitation of existing technologies and the exploration of new technologies from a value chain perspective.

8.2. Implications for Practice

This research also provides valuable insights for managers and policymakers on how to manage and regulate fast-evolving and highly costly technologies in value chains. Specifically, the arrival of 5G technology grants greater importance to MVNOs in the telecommunication market (Grimaldo 2019). Managers need to evaluate market expansion strategies involving overlapping technology markets in terms of value chain governance between internal resource commitment and delegation. Managers should, at the onset of a new technology, proactively access new markets by committing internal assets, which promotes learning and innovation. However, they should also leverage complementary resources from independent third parties to continue to grow with existing, mature technologies.

Further, our findings are also relevant to other high-tech industries with fast clockspeeds, where the challenge from overlapping generations of technologies is inevitable. For instance, the industry of graphic processing units (GPU) usually releases new generations of

products with radical and disruptive technologies every few years. Nvidia, the largest GPU designer worldwide, has released GTX 10, 20, and 30 series GPU in the past decade, where each generation offers major performance boosts nearly double the previous generation (Chen 2022). This rapid architectural and technological progression poses a constant challenge for Nvidia to continuously exploit current technologies while exploring new ones. Also similar to MNOs, Nvidia directly sells to consumers via PC manufacturers (e.g., Dell). To capture specific market segments, such as the high-end DIY market, Nvidia markets independently packaged GPUs directly to consumers and, at the same time, delegates to third-party resellers who complement Nvidia in technology, service, and products (e.g., better performance-tuning software or cooling and lighting solutions). The findings of this study can be helpful for managers in other high-tech industries, such as GPU, to address the challenge of technological dynamics with value chain solutions.

8.3. Limitations and Future Research

Our study is not without limitations. While our unique data has led to the first causal study on governance and overlapping generations of technologies in value chains, the study would have provided a more in-depth contribution with MVNO-level subscription information. However, MVNOs are mostly private firms, whose financial and operating performance information is not available. Future research can examine the impact of MVNOs from a value chain perspective and compare the performance of MNOs and MVNOs as well as consumer welfare. Second, our study does not cover detailed information on the service plans and pricing mechanisms of MVNOs. Future research could enhance our results by including MVNO-level service plans and pricing strategies, or even comparing these with MNO plans to determine the optimal differentiation levels for MVNOs.

Endnotes

¹ 2G or 3G is based on a set of technological standards specified by the International Telecommunication Union (ITU). Each generation of technology contains variants with improvements. We aggregate the variants into broadly recognized generation levels for simplicity.

² See discussion in Section 6.2 and technical details in Online Appendix E.4.

³ For example, T-Mobile offers services in the U.S. Virgin Islands (an external U.S. territory). However, the T-Mobile operations in the U.S. Virgin Islands are parallel to those in the U.S. mainland, and both are operated by T-Mobile US. Therefore, T-Mobile Virgin Islands does not make separate decisions on value chain structure, so it would likely contaminate our causal inferences if included as an independent MNO in the sample. The results using the sample combining MNOs in non-self-governing territories and governing countries are consistent with our main findings (see Online Appendix I).

⁴ In the matching algorithm of dPSM, subjects selected as controls in the previous periods are removed from the candidate control group in future matching. Thus, by constructing two samples, we are able to improve the quality of matching by slightly increasing the size of the candidate control group over time. The estimation results using one sample are highly consistent with the results using two samples.

⁵ Note that the reduction in the number of MVNO launches is due to differences in the number of MVNO launches across countries. In the robustness test section, we test the higher matching ratio with a greater sample size and the results are still consistent (see Section 6.2).

⁶ Since we use 2G and 3G market shares to estimate the treatment effects, an increase in the market share of an MNO would most likely lead to a decrease in the other MNOs' market shares in the same market. Thus, matching treated MNOs with control MNOs in other countries mitigates such zero-sum situations.

⁷ Most countries have a limited number of MNOs, and thus "same-country matching" leaves a limited number of candidate MNOs for the matching algorithm. As a result, matching in the same country significantly reduces the matching quality and thus dilutes the causal estimation process.

⁸ Note that the numbers of MNOs in the constructed treatment and control groups are different from the exploratory analysis because we apply a more strict and rigorous matching algorithm.

⁹ Alternative matching variables are tested in Section 6.2.

¹⁰ See the sliding-window adjustment example in Online Appendix A.

¹¹ We also run a sequential matching model and a multinomial logit model, which separate the scenarios of launching an MVNO from launching a branded or third-party MVNO. The results are qualitatively consistent (see Online Appendix G).

¹² Given the complexity of dPSM, Figure 6 shows an example with an identified treatment subject in a given period.

¹³ The results are consistent with shorter and longer post-treatment windows ($k_e = 2$ and $k_e = 8$). See Section 6.2 for details.

¹⁴ We use years since 2G/3G technology launch as an alternative measure of maturity and our results still hold (see Online Appendix J).

¹⁵ We thank the Department Editor, the Associate Editor, and an anonymous reviewer for this suggestion.

¹⁶ See <https://data.worldbank.org/>.

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